Revisiting decision trees

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Decision trees can offer valuable insights into the structure of a strategic decision problem

Introduction

Decision trees have featured in the decision analysis and management decision-making textbooks for many years e.g. Bierman[1], Curwin and Slater[2], Gregory[3], Lucey[4] and Thirkettle and Wheldon[5], yet personal experience and experience of colleagues working with managers and students on management programmes would strongly suggest that students and managers find decision trees to be of limited value in modelling the decision process. Based on evidence from focused group discussions with students of management studies, conducted over a period of several years, this article explores the hypothesis that there are three significant and interlinked reasons why managers do not make effective use of this analysis technique:

H1: Students and managers have difficulty with some of the probabilistic concepts inherent in decision trees and decision making under uncertainty.

H2: Students and managers cannot translate the decision tree model which is still essentially presented in the textbooks as a paper-based model, into a spreadsheet environment that allows easier manipulation of numbers.

H3: Students and managers do not make decisions in a formal way that takes the long-term perspective inherent in the decision tree approach.

This article demonstrates approaches for alleviating the first two hypotheses and considers the significance of the third hypothesis. The article starts with a basic review of the structure of a decision tree and introduces an example which can be used to illustrate some important issues.

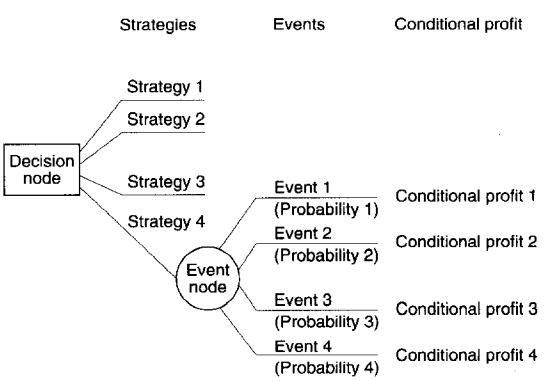
Decision trees

Decision trees are designed to assist the decision maker with "longitudinal" decision making, where, for example, a decision may be made, some further events may occur, a subsequent decision may be necessary and yet more events may occur. They are a tool which should assist the decision maker when making an initial decision to consider the range of events and subsequent decisions that may occur or be necessary and should encourage the decision maker to think beyond the immediate decision. The series of events and decisions may occur in a relatively compressed period of time, or may be extended over a long period, of say, several years. Typically, then, decision trees may be used for forecasting. All forecasting is conducted under uncertainty, or, in other words, we do not know what will happen next! We return to this issue later.

In order to appreciate the comments later in this article it is useful to revisit the basic structure of a decision tree. Figure 1 shows the basic structure of a decision tree.

For example, a company is trying to decide the size of a new plant. At present the company has only a minimal sales effort, but when the new plant is completed a major promotion effort will be undertaken. Management is somewhat uncertain about the likely success of this effort. It is estimated that there is a 0.3 chance that the company will capture a significant market share, and a 0.7 chance that it will capture a moderate market share. A larger plant will be necessary if a significant market share is captured. A small plant would suffice if the market share

Figure 1. *The structure of a decision tree*



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were only moderate. The large plant costs £8 million, the small plant costs £5 million.

If a significant market share emerges, the estimated value of resulting profits (excluding the cost of the plant) is £14 million; if a moderate share is achieved, the present value of the resulting profits is £9 million. Management has one other alternative. They could build a small plant, wait to see the level of sales, and then expand the plant if the situation warrants. It would cost an additional £4.5 million to expand the small plant (see Figure 2).

Decision trees are constructed by a two-step process:

(1) forward pass; followed by

(2) backward pass.

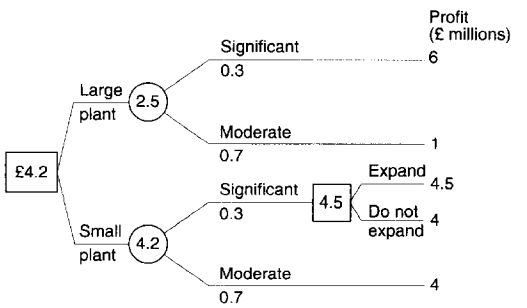
The forward pass involves the decision maker in identifying the decisions to be made, the events that might occur and the sequence of decisions and events. The forward pass reveals the structure of the problem. During the forward pass, conditional profits must be calculated (i.e. the profits that will be achieved if certain strategies are adopted and certain events occur) and the probability of events must be assessed.

The backward pass is concerned with analysing the decision problem. Expected values are estimated by working backwards through the tree, as follows:

- For each set of event branches the expected value is calculated.
- For each set of decision branches the one with the highest expected value is selected.
- The strategy with the highest expected value is selected and \\ is drawn through lines representing the other options.

The final analysis should lead to a decision. Once this decision has been identified, a preferred strategy will emerge.

Figure 2. *An example of a decision tree*



Using decision trees for real situations

A very real problem in constructing decision trees for real business decisions is the uncertainty associated with many elements of the decision model. A decision tree is supposed to cater for uncertainty of events and outcomes by considering all possible events and allocating probabilities to the likelihood of their occurring. This is far too simplistic an approach to uncertainty in the longitudinal decision-making process, because uncertainty is also embedded in the model in a number of other respects:

- There is uncertainty concerning whether all possible events, decisions and their timing have been correctly identified.
- There is uncertainty about the values of the probabilities, particularly where conditional probabilities need to be estimated.
- Different decision makers may apply different decision criteria, and thereby arrive at different outcomes.
- It can be difficult to estimate conditional profits or other values of other performance criteria.

Many of these uncertainties could be reflected in a more complex probabilistic model, but the models are likely to be sufficiently complex to deter all but the most quantitatively-minded manager.

It is at this point that many students and managers reject decision trees as having no practical application and thereby reject a very valuable tool for extending their decision-making horizons. So, of what value are decision trees?

Making decision trees useful

The way forward is to recognize that decision trees are a useful way to structure the longitudinal decision-making process. Even a simple decision tree with one decision, followed by a set of events and another decision with its set of events, is a very useful model. This section explores some of the different ways in which decision trees can be used to support the decision-making process.

Using the forward pass alone

Operational and tactical decisions are often made without taking a longer-term view, but it is imperative that strategic decision making considers a longitudinal sequence of decisions. Managers find it very difficult to perform such an analysis without some kind of representation, such as a decision tree, which allows them to identify and consider a range of strategies and potential events. The forward pass in a decision tree can then be used as a means of thinking about the strategies

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available. For example, when selecting a computer system including hardware, software and telecommunications links, it may at first appear that the decision is simply a once-and-for-all selection of the package offered by a vendor in preference to those offered by other vendors. However, in reality this is rarely the case. Normally the decision is made in some uncertainty about future system requirements and future technological development.

A decision tree can be a considerable aid in cultivating a systematic approach

For example, in the future there may be a release of a new version of the software package. This might lead to the need to upgrade the memory in the hardware platform in the following year or two. The decision maker has the option of purchasing hardware that already has sufficient memory for the potential software upgrade in advance of demand for the additional memory. The display of such strategies and events in the form of a decision tree can be a considerable aid in cultivating a systematic approach to the longer-term perspective.

Payoffs

Conventionally payoffs in decision trees are displayed in the final column and often depict profit. It is important to remember that other objectives might lead to alternative payoffs. For example, in a project where cost minimization was central the payoffs might show costs, with the objective being to minimize costs. Alternatively other performance indicators associated with numbers of service encounters or quantity of stock inventory might also be used in the payoff column.

Decision criteria

The standard approach to decision trees often uses the expected value decision criteria. It is important to recognize that this embodies a specific approach to decision making. The expected value decision criterion uses all of the information concerning the consequences of each event, and how likely each event is to occur. It does not, however, take into account the decision maker's attitude to risk. Accordingly the criterion is most appropriate when risk associated with a decision is not high.

There are several alternative decision criteria.

Equally likely criterion

This assumes that each of the potential events is equally likely. Under this assumption the possible consequences of each decision are added, and the total is divided by the total number of possible events. The strategy with the highest value is deemed to be the most desirable.

This is a sensible strategy if we have no knowledge of the relative probability of each event happening. If we do have such information, then we should use it!

Maximax

When applying the maximax strategy we look for the largest possible profit that can be achieved for each strategy, and choose the strategy with the highest potential profit. Often, in practice, but not always, this strategy also has the highest possible loss associated with it. This is regarded as a go-for-it strategy, and is not prudent. Such a strategy would only be adopted by a manager who was in a position to accept high risks. In a situation where there was no uncertainty as to which event might happen, if we knew that a specific event would happen, then we would always opt for the strategy that led to the maximum profit. Uncertainty introduces risk, which means that we might choose to be more conservative.

Minimax

The minimax criterion suggests that we choose the act with the smallest maximum possible loss, or alternatively, with the largest minimum profit. We evaluate the strategies under this criterion. Unfortunately, in practice, this criterion tends to lead to a decision to do nothing, and leads to conservatism and stagnation. A decision which involves an element of risk is likely to have potential for greater success.

Maximum likelihood criterion

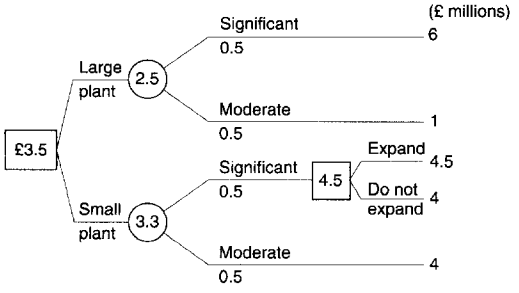
The maximum likelihood criterion leads the decision maker to choose the event that is considered to be most likely, and then to choose the best decision for that event. This approach fails to use much of the available information (i.e. that concerning the less likely events) and can therefore lead to unreasonable decisions. It is, however, a criterion often used in practice by managers when they shrink from a systematic evaluation of all possible events. They start by asking themselves what is most likely to happen, and then make their decision on the basis of this prediction. This can often provide a way forward when there is insufficient information, but it does offer a limited perspective.

The application of different decision criteria in a decision tree will often suggest:

- a different expected value at the first decision node;
- the adoption of a different strategy.

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Figure 3. *Decision tree using the equally likely decision criterion*



The application of a different decision criterion is a stark reminder that the most appropriate decision strategy depends on the decision maker's attitude to risk.

The decision tree in Figure 2 uses the expected value decision criterion. If we were to apply the equally likely decision criterion the decision tree would appear as in Figure 3. This would yield a different anticipated value at the decision node. Application of the maximum likelihood decision criterion would lead to a strategy which ignored the possibility that there might be a need to expand in the future.

Sensitivity analysis demonstrates what happens if changes are made in one of the decision or exogenous

Figure 4. *A decision tree built using a spreadsheet*

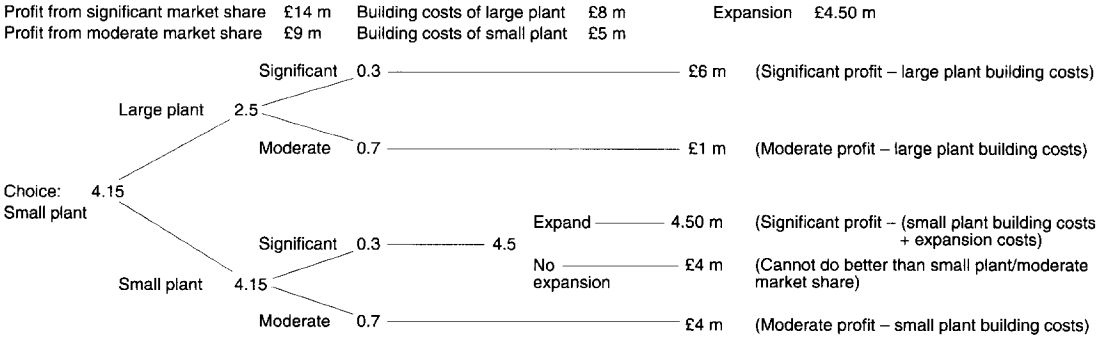


Figure 5. *Sensitivity analysis*

Chance of significant market share	Profit	Decision
	4.15	Small plant
0.1	4.05	Small plant
0.2	4.1	Small plant
0.3	4.15	Small plant
0.4	4.2	Small plant
0.5	4.25	Small plant

Even with equal likelihood, 0.5 chance the choice from the decision tree is to build a small plant

A two-way sensitivity analysis of different levels of profit

		Significant market share (£m)				
	£4.15	£12.00	£13.00	£14.00	£15.00	£16.00
Moderate market share (£m)	£8.00	£3.00	£3.15	£3.45	£3.75	£4.05
	£8.50	£3.50	£3.50	£3.80	£4.10	£4.40
	£9.00	£4.00	£4.00	£4.15	£4.45	£4.75
	£9.50	£4.50	£4.50	£4.50	£4.80	£5.10
	£10.00	£5.00	£5.00	£5.00	£5.15	£5.45

Keeping the chance of significant market share at 0.3 this analysis investigates the influence that varying estimates of profits may have on the decision process.

		Significant market share (£m)				
	Small plant	£12.00	£13.00	£14.00	£15.00	£16.00
Moderate market share	£8.00	Small plant	Small plant	Small plant	Small plant	Small plant
	£8.50	Small plant	Small plant	Small plant	Small plant	Small plant
	£9.00	Small plant	Small plant	Small plant	Small plant	Small plant
	£9.50	Small plant	Small plant	Small plant	Small plant	Small plant
	£10.00	Small plant	Small plant	Small plant	Small plant	Small plant

The upper table is the expected value of profit and the lower table the associated decision

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variables. By exploring the effect of such changes it is possible to judge how sensitive the profit is to change in specific variables. Sensitivity analysis may be applied on decision trees in respect of the values of variables such as pay-offs, costs incurred by successive decisions and the probabilities of events in the decision tree. This allows the effect of variations in the values of variables to be investigated. The easiest way to undertake a sensitivity analysis is by creating a decision tree on a spreadsheet and using some of the sensitivity analysis tools to record the outcomes of variations in specific variables. The example in Figures 4 and 5 demonstrates how this might be achieved.

Conclusion

This article has explored, using appropriate examples, the ways in which decision trees can be used by the manager to assist in the longitudinal decision-making process. Because of the mathematical concepts associated with complex decision trees, managers can be reluctant to attempt to use decision tree models. A recognition that such models can be simply developed in a spreadsheet environment, and can then be used for sensitivity analysis using different decision criteria, demonstrates that decision trees can offer valuable insights into the structure of a strategic decision problem.

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Application questions

(1) Evaluate the usefulness of mechanistic models such as decision trees. What other decision-making approaches are there?

(2) How effectively does the decision tree analysis work with imperfect forecast information?

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